SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes
Total Marks:20

QUESTIONS: 2

Annexure A

Descriptive Written Test

Code of Conduct:

I M. Gokul (192565067) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

level-3

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Descriptive written test.

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REG NO : 192565067

COURSE : Artificial intelligence

CODE : CSA1714.

SLOT : Slot - D

Smart Home Automation System - Intelligent Agents

Introduction :

A Smart home automation system uses Artificial intelligence (AI) to automatically Control lighting, temperature, and security according to user behavior and environmental Conditions. The system receives inputs from Sensors Such as motion detectors, temperature sensors.

1. Simple Reflex agent

A simple Reflex Agent makes decisions only based on the current input. it follows predefined condition action rules without considering previous events.

working in the Smart Home:

* Turns on lights when motion is detected.
* Turns off lights when no movement is detected
* Switches on the air Conditioner if the temperature exceeds a fixed limit
* Activates the alarm when a door Sensor detects intrusion.

Advantages:

* very fast response
* easy to design and Implement
* Requires less memory.

Example:
It different path,

Disadvantages
* Cannot learn user preferences,
* cannot remember previous events.
* Performs poorly in Complex situations.

2. Model - Based Reflex agent

Working in the Smart Home
* Remembers whether a room was recently occupied.
* Keeps track of indoor temperature changes
* understands whether doors and windows are open or closed.
* Makes decisions based on both Current and past information.

Advantages:
* Handles partially observable environments.
* Makes more accurate decisions
* Better than Simple reflex agents

Disadvantages:
* Requires additional memory
* More Complex to implement.

Conclusion:

Although Simple Reflex Agents are Suitable for basic automation, modern Smart homes require intelligent decision - making. A hybrid Model- Based + Goal - Based + utility -Based Agent provides the best Solution because it learns user preferences, adapts to environmental changes improves Comfort, Saves energy, and enhances home Security.

2. Robot in an unknown Maze - uninformed Search Strategies (5)

Introduction:

A robot is placed in an unknown maze and must find the exit without prior knowledge of the environment. Since the robot does not know the layout, it uses uninformed (blind) search algorithms, which explore the maze without heuristic information.

Two commonly used uninformed search algorithms are:

* uniform cost search (UCS)
* Iterative Deepening Search (IDS)

1. uniform cost Search (UCS)

uniform cost search expands the node with the lowest cumulative path cost first.

working in the Maze:

* The robot begins at the starting position.
* It explores paths based on the total travel cost.
* The lowest-cost path is always expanded first.
* The process continues until the exit is found.